

A Gaussian Thermal Reference and Independent Memory-Kernel Audit for MQM Capture

Ben Mayes

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Abstract—We specify the finite-temperature Gaussian continuum of the MQM flat-band latch, preserving correlated bath covariance. A coherent-displacement argument bounds thermal-to-vacuum retained channel distance by $t^2 V_\beta$. The bound is inconclusive at the requested inverse temperature $\beta = 5/16$, but a separately frozen cooling design $\beta = 7/16$ has a sufficient retained error ceiling below 2^{-16} at $t = 16$. An independent Volterra solver reproduces the vacuum continuum result within 1.53×10^{-10} . Thermal quadrature refinement is recorded without modifying the spectrum, tolerance, or hypotheses. No direct thermal channel simulation, hardware validation, permanent hold, or complete-word certificate is claimed.

Index Terms—Quantum control, thermal reservoirs, Gaussian states, non-Markovian dynamics, channel bounds.

I. SOURCE MODEL AND FROZEN EVALUATION

The Mayes Quantum Method (MQM) source is the same compact flat spectrum $J(\omega) = \kappa/(2\pi)$ on $\omega \in [\Omega - B, \Omega + B]$, with $\Omega = 64$, $B = 16$, $\kappa = 8$, and port couplings $h = g = 1$, $\lambda^2 = 4\sqrt{2}$. Fixed launch is the admitted $pR0$ state; the bank and arbitrary reference are identity spectators. The inherited vacuum certificate is

$$E_0 = \frac{3}{262144} + 2^{-128}, \quad q_0(16) < E_0. \quad (1)$$

The primary thermal request remains $\beta = 5/16$. Before thermal evaluation we also freeze exploratory cooling values $3/8$, $7/16$, and $1/2$, retaining the same spectrum and deadline. A sufficient colder point is not a replacement for the original-temperature target or a measured thermal threshold.

Thermal absorption accesses the full eight-dimensional admitted clock-auxiliary sector, including charge-two states $pR1, mR1, eR1$. The four live-state vacuum restriction cannot be reused for thermal propagation. Let $S = I_C \otimes \sigma_R^-$, with

$$H = H_S + \int \omega b_\omega^\dagger b_\omega d\omega + S^\dagger F + S F^\dagger, \quad (2)$$

$$F = \int \sqrt{J(\omega)} b_\omega d\omega. \quad (3)$$

The initial bath is the stationary Gaussian equilibrium state of the *free* bath, factorized from the prepared system launch. Coupled equilibrium preparation is not assumed. The standard Gaussian continuum reference is defined through finite-mode regularizations and their reduced dynamics; existence of this reference limit is an explicit model assumption, not a material validation.

II. CORRELATIONS AND THERMOFIELD SPECIFICATION

Define $n_\beta(\omega) = (e^{\beta\omega} - 1)^{-1}$. The exact bath correlations are

$$\langle F(t)F^\dagger(0) \rangle_\beta = \int J(\omega)[1 + n_\beta(\omega)]e^{-i\omega t} d\omega, \quad (4)$$

$$\langle F^\dagger(t)F(0) \rangle_\beta = \int J(\omega)n_\beta(\omega)e^{+i\omega t} d\omega. \quad (5)$$

For the normalized Legendre functions $\phi_n(\delta)$ on $[-B, B]$, the transformed covariance is

$$N_{nm} = \int_{-B}^B \phi_n(\delta)\phi_m(\delta)n_\beta(\Omega + \delta) d\delta. \quad (6)$$

It is generally non-diagonal. Independent thermal preparation of chain sites changes the source model. The degrees 0–7 covariance projection at $\beta = 5/16$ has a largest off-diagonal entry in magnitude 5.78×10^{-8} ; quadratures of orders 128 and 256 agree within 8.1×10^{-21} . These numerical checks do not simulate a thermal channel.

An equivalent thermofield specification uses independent vacuum fields c_ω, d_ω and

$$F_\beta = \int \sqrt{J(\omega)}[\sqrt{1 + n_\beta} c_\omega + \sqrt{n_\beta} d_\omega^\dagger] d\omega, \quad (7)$$

with auxiliary free generator $\int \omega(c_\omega^\dagger c_\omega - d_\omega^\dagger d_\omega) d\omega$. This representation is prior art [1]. The auxiliary negative energies are purification bookkeeping, not a proposed physical heat source. The upward term $S^\dagger d_\omega^\dagger$ creates a system and auxiliary excitation together. It preserves signed charge $N_S + N_c - N_d$, not positive total occupation. Multiphoton processes cannot be discarded by invoking the vacuum sector theorem.

III. A THERMAL CHANNEL DEBIT WITHOUT PHOTON TRUNCATION

Proposition 1. For the declared exchange coupling, let

$$V_\beta = \int J(\omega)n_\beta(\omega) d\omega. \quad (8)$$

The retained thermal and vacuum channels obey

$$\frac{1}{2} \|\Phi_t^\beta - \Phi_t^0\|_\diamond \leq \min\{1, t^2 V_\beta\}. \quad (9)$$

Proof. Each finite-mode thermal state has the exact positive coherent-state mixture

$$\rho_{\text{th}}(\bar{n}) = \int \frac{d^2\alpha}{\pi\bar{n}} e^{-|\alpha|^2/\bar{n}} |\alpha\rangle\langle\alpha|. \quad (10)$$

TABLE I
SUFFICIENT THERMAL ERROR BUDGETS AT $t = 16$

β	Thermal debit ceiling	Total ceiling, approx.	Status
5/16	16384/31457265	5.323×10^{-4}	Open
3/8	8192/301989879	3.857×10^{-5}	Open
7/16	16384/22548578283	1.217×10^{-5}	Sufficient
1/2	2048/51539607549	1.148×10^{-5}	Sufficient

Gaussian integration gives number-state coefficient $\bar{n}^j/(1 + \bar{n})^{j+1}$, equal to the thermal geometric distribution. Time-dependent bath displacement converts a coherent launch into vacuum dynamics plus the drive $D_\alpha(t) = \zeta(t)S^\dagger + \zeta^*(t)S$. Final bath displacements disappear under trace. This is a proof representation, not an externally scheduled control requirement.

The centered circular Gaussian field satisfies $\mathbb{E}\zeta = 0$ and $\mathbb{E}|\zeta(t)|^2 = V_\beta$. Since S is an exact auxiliary-qubit lowering operator, $\|D_\alpha(t)\| = |\zeta(t)|$. Conjugation into the vacuum interaction picture preserves this norm; the commutator superoperator has diamond norm at most $2|\zeta(t)|$. The mean first-order Duhamel term vanishes. For the second remainder, unitary channel norm is one and Cauchy–Schwarz gives $\mathbb{E}|\zeta(s)\zeta(u)| \leq V_\beta$. Therefore

$$\frac{1}{2}\|\mathbb{E}\Phi_\alpha(t) - \Phi_0(t)\|_\diamond \leq 2 \int_0^t ds \int_0^s du V_\beta = t^2 V_\beta. \quad (11)$$

Partial trace contracts the bound. Taking the declared Gaussian continuum regularization preserves the finite-mode inequality. This argument bounds the complete retained channel and includes thermal occupation through the Gaussian mixture; it is not a first-photon truncation or a Born/Markov approximation.

For this flat band,

$$V_\beta = \frac{\kappa}{2\pi\beta} \log \frac{1 - e^{-\beta(\Omega+B)}}{1 - e^{-\beta(\Omega-B)}}. \quad (12)$$

A convenient rational allowance follows from

$$t^2 V_\beta \leq \frac{\kappa t^2}{2\pi\beta} \frac{e^{-\beta(\Omega-B)}}{1 - e^{-\beta(\Omega-B)}}. \quad (13)$$

Positive degree-80 Taylor lower sums establish $e^{15} > 2^{21}$, $e^{18} > 2^{25}$, $e^{21} > 2^{30}$, and $e^{24} > 2^{34}$. Using $\pi > 3$ gives Table I. “Open” means the upper bound is inconclusive, not that thermal operation fails. For the independently declared $\beta = 7/16$ design, triangle composition with the inherited vacuum certificate yields

$$E_\beta < E_0 + \frac{16384}{21(2^{30} - 1)} < 2^{-16}. \quad (14)$$

This is a derived sufficient bound, not a simulated thermal error. At fixed energy scale the temperature ratio relative to $\beta = 5/16$ is $5/7$; no kelvin assignment or physical cooling achievement is asserted.

IV. INDEPENDENT VACUUM MEMORY-KERNEL CHECK

Eliminating the vacuum flat bath directly gives the Volterra equation

$$\dot{v} = A_0 v - e_3 \int_0^t C_0(t-s) v_3(s) ds, \quad (15)$$

where A_0 is the real skew four-state path with weights $(1, 1, \lambda)$, and

$$C_0(s) = \frac{8 \sin(16s)}{\pi s}, \quad C_0(0) = 128/\pi. \quad (16)$$

An implicit trapezoidal ODE step and trapezoidal memory convolution use no bath-chain matrix or matrix exponential. Frozen steps $1/256$, $1/512$, and $1/1024$ give $q(16)$ equal to 1.07101055 , 1.07119391 , and 1.07123975 , respectively, times 10^{-5} . The finest value differs from the parent directed interval by 1.53×10^{-10} ; the last-two difference is 4.59×10^{-10} , meeting the frozen 10^{-8} criterion. This is a separate numerical formulation, not another directed theorem or independent external review.

The thermal kernel can also be checked without chain coordinates. Let $C_n(s) = \int J(\Omega + \delta) n_\beta(\Omega + \delta) e^{-i\delta s} d\delta$. Its Bose-series form is

$$C_n(s) = \frac{\kappa}{2\pi} \sum_{\ell \geq 1} \frac{e^{-\ell\beta(\Omega-B)} e^{iBs} - e^{-\ell\beta(\Omega+B)} e^{-iBs}}{\ell\beta + is}. \quad (17)$$

Then the rotating downward kernel is $C_0 + C_n$; the upward correlation uses the conjugate and opposite laboratory carrier phase. A four-term series has an analytic omitted-tail allowance below 1.1×10^{-31} at primary $\beta = 5/16$. Frequency quadrature confirms the kernel at $s = 0, 1/16, 1, 16$.

V. NUMERICAL REPAIR AND PROMOTION BOUNDARY

The frozen order-128/256 thermal quadrature failed at $s = 16$, with discrepancy 2.03×10^{-8} . Orders 256/512 reduce the discrepancy below 7.0×10^{-20} . The original recipe and failure diagnostic are preserved; only resolution changed, with the same spectrum, time, and 10^{-12} tolerance. Covariance quadrature and the vacuum Volterra solver required no repair.

The thermal reference construction and sufficient colder-point bound are conditional mathematical evidence. The requested $\beta = 5/16$ target remains open. No finite-temperature trajectory was simulated. The former vacuum one-excitation norm ceiling cannot be applied to thermofield multiphoton dynamics. This release does not close the previous Ohmic family, the thirteen-step microscopic word, permanent retention, reservoir refresh, native admission, physical two-local compilation, or hardware advantage. Physical promotion remains zero.

REFERENCES

- [1] I. de Vega and M.-C. Bañuls, “Thermofield-based chain-mapping approach for open quantum systems,” *Phys. Rev. A*, vol. 92, p. 052116, 2015.